

Redes Móveis e Internet das Coisas (Mobile Networks and IoT)

3rd Quarter (2025/2026)

Instituto Superior Técnico / Universidade de Lisboa

Prof. António Grilo

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Contact

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- Miguel Andrade (**lab**)
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Classes

- Lectures:
 - Tuesday 10:00-12:00 QA02.1
 - Friday 10:00-12:00 EA1
- Lab:
 - RMICL02 (Wed 10:30 - 12:00 - LSD2 ; Tue 13:30 - 15:00 - LSD2)
 - RMICL03 (Thu 13:00 - 14:00 - LSD2 ; Wed 14:30 - 16:00 - LSD2)
 - RMICL04 (Fri 14:00 - 15:30 - LSD2 ; Thu 11:30 - 13:00 - LSD2)
 - RMICL05 (Thu 14:30 - 16:00 - LSD2 ; Fri 12:00 - 13:30 - LSD2)
- Office Hours:
 - António Grilo (Teams): Monday 14:00-18:00
 - Miguel Andrade (Teams): Mon 09:00-11:00

Evaluation

- Components:

- Exam (E): 50%
- Lab Assignments (L): 10%
 - 3 weeks
 - Android+Cloud+WiFi (1)
 - LoRaWAN (1)
 - NB-IoT (1)
- Project (P): 40%
 - 4 weeks

- Final grade:

- $EN: 0.1L + 0.4P + 0.5E$
- $ER: 0.4P + 0.6E$
- Minimum grade P: 8,0
- Minimum grade E: 8,0

Important Dates

- *Exam:*
 - *Normal Season: 14/Apr 13:00*
 - *Recovery Season: 30/Jun 10:30*
 - *Special Season: ?*
- *Lab Assignments:*
 - *Assignment 1: week 1*
 - *Assignment 2: week 2*
 - *Assignment 3: week 3*
- *Project:*
 - *Weeks 4, 5, 6 and 7*
 - *Delivery: 05/Apr*
 - *Submitted through Fenix*
 - *Demo+Discussion: 06/Apr – 10/Apr*

Booking and Delivery (Lab)

- Lab groups:
 - Groups of two students.
 - Students must be registered in the same shift.
- Lab assignment evaluation:
 - Functionality check in the classroom

Projects: Booking and Delivery

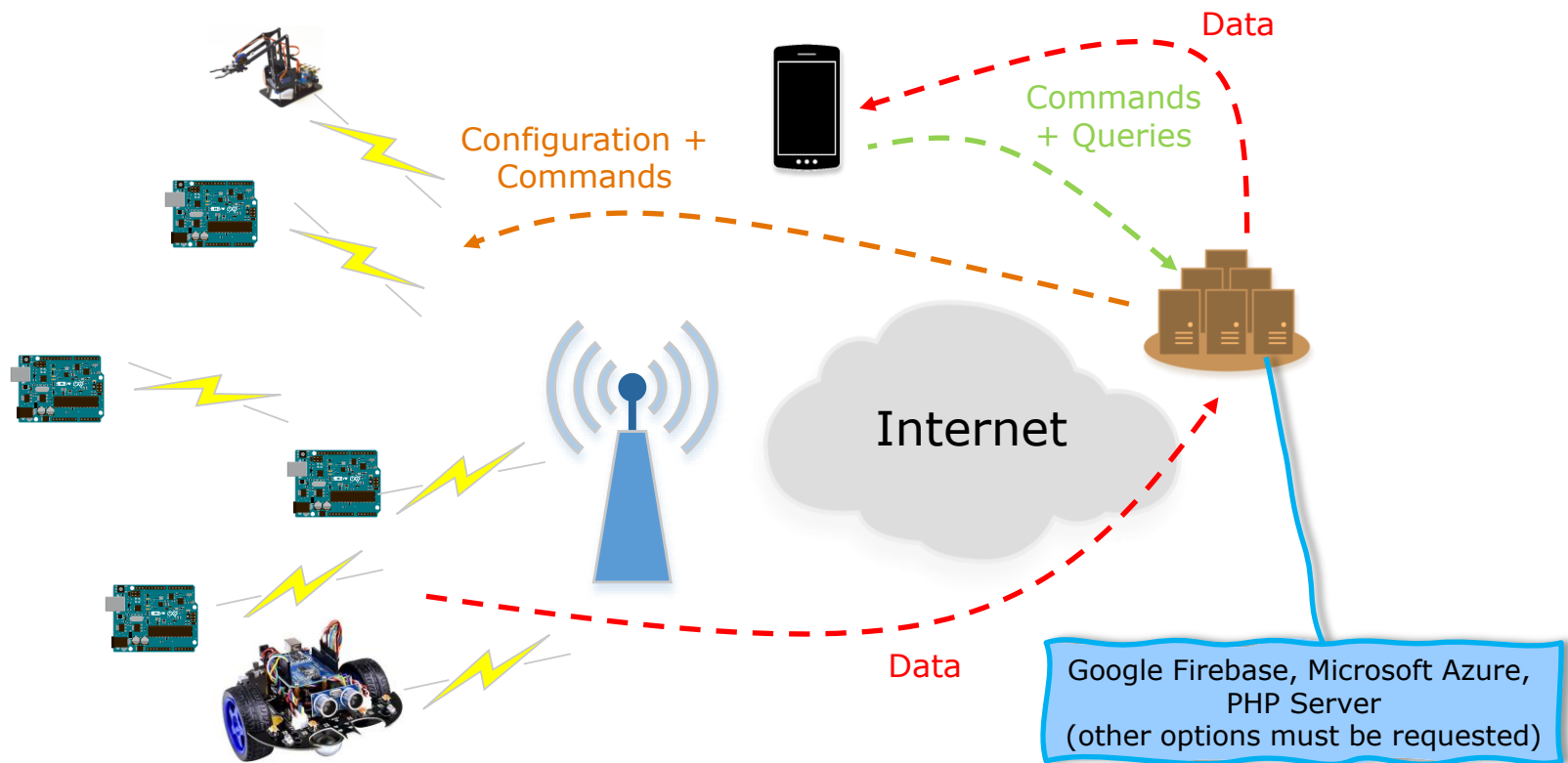
- Topics: see next slide
- Project Selection
 - Formulary submitted through Fenix
 - Details will be provided later
- Project assignment policy:
 - Only a limited number of groups can choose each project.
 - FIFO assignment policy.
 - Subject to inventory constraints
- Project delivery: Fenix

Projects: Topics and Requirements

- Overall topic: Internet of Things
- Specific topic: Proposals listed in the course's webpage. Topics proposed by the students may be accepted (must be discussed with the professor).
- Requirements must be adapted to the available material (listed in the course's webpage), or material brought by the students.

Projects: Basic Requirements and Architecture

- Project must close the loop between sensors/actuators, webserver and mobile App.
- Project must use WiFi, LoRaWAN and/or NB-IoT communication. In addition, may also use Bluetooth and/or IEEE 802.15.4.



Projects: Basic Requirements and Architecture

- Project must close the loop between sensors/actuators, webserver and mobile App.
- Project must use WiFi, LoRaWAN and/or NB-IoT communication. In addition, it may also use Bluetooth and/or IEEE 802.15.4.
- Minimal configurations:
 - 2 simple sensors + 1 simple actuator + 1 complex sensor
 - 3 simple sensors + 1 complex actuator
 - 1 complex sensor + 1 simple sensor + 1 simple actuator + machine learning
 - 1 robot + 1 complex sensor + 1 simple sensor
 - 1 robot + 1 complex sensor + machine learning
 - 1 robot + Bluetooth/IEEE 802.15.4 + 1 simple sensor
 - Other configurations considered similar in complexity

Projects List (suggested)

- 1. Monitoring water quality (e.g., turbidity, water flow, and temperature) in fish farm pond.**
- 2. Smart wine cellar (control of temperature, wine quantity, gases (CO₂, O₂), etc.).**
- 3. Pet monitoring and entertainment.**
- 4. Reconnaissance of indoor spaces in support of police operations (robot or drone).**
- 5. Drone-based Data Mobile Ubiquitous LAN Extension (MULE) to gather measurements from sensors located in distant or hazardous areas.**
- 6. Power transformer damage prevention through temperature monitoring.**
- 7. Smart agriculture based on local weather station and other sensors.**
- 8. Fire detection and extinction in buildings.**
- 9. Bodycam and biometric sensors for firefighters or policemen.**
- 10. Access control in critical facilities.**
- 11. Smart waste monitoring solution for Smart Cities.**

- 12. More to come... Bring your own idea! We are also open to analyze student project proposals, as long as they respect minimum requirements.**

- *LE3*
- *Open to students of AC Comp courses*
- *Mentored by Hacker School members*
- *Can provide HW and SW support*
- *Electronics Lab*
- *3D Printing*

